Lecture 1 – Basic Networking Concepts

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1. Introduction to Computer Networks

Definition of a Network

A **computer network** is a collection of interconnected devices (computers, routers, servers, etc.) that communicate with each other using communication links.

Key Characteristics

- Devices exchange **data**.
- Communication occurs via **wired or wireless media**.
- Governed by **protocols**.

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2. The Internet

Definition

The **Internet** is a **global network of interconnected networks** that communicate using standardized protocols.

Key Components

1. **Hosts (End Systems)**
    - Computers, smartphones, IoT devices
    - Run applications
    - Located at the **network edge**
2. **Communication Links**
    - Connect devices together
    - Examples:
        - Fiber optic
        - Copper cables
        - Wireless
        - Satellite

3. **Packet Switches**
    - Forward data packets
    - Examples:
        - Routers
        - Switches
4. **Internet Service Providers (ISPs)**
    - Organizations that provide Internet connectivity.

Internet Structure

The Internet consists of:

Network Edge

Where **end devices (hosts)** are located.

Examples:

- PCs
- Mobile phones
- Servers

Servers are often located in **data centers**.

Network Core

The **central part of the network** that connects edge devices.

Consists of:

- Routers
- High-speed links

Responsible for:

- **Routing**
- **Forwarding packets**

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3. Internet as Infrastructure and Service

Internet as Infrastructure

Provides services for applications such as:

- Web browsing
- Email
- Video streaming
- Online gaming
- Social media
- Cloud computing

Internet as a Service Platform

Provides an **API-like interface** for applications.

Applications use the Internet through:

- **Transport services**
- **Protocols**

Example services:

- HTTP
- TCP
- IP
- WiFi

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4. Protocols

Definition

A **protocol** is a set of rules that governs communication between devices.

Protocols define:

1. **Message format**
2. **Order of messages**
3. **Actions taken when messages are received**

Example (Human Protocol)

Conversation:

1. Person A: "What time is it?"
2. Person B: "2:00"

Rules:

- Greeting
- Question
- Response

Example (Network Protocol)

Example communication:

1. Client sends **TCP connection request**
2. Server responds
3. Client sends **HTTP request**

Examples of Internet Protocols

- **HTTP** – Web communication
- **TCP** – Reliable transport
- **IP** – Addressing and routing
- **WiFi** – Wireless communication
- **Ethernet** – Wired LAN communication

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5. Data Communication

Definition

Data communication is the exchange of data between devices using a transmission medium.

Effectiveness of Data Communication

Three main factors:

1. Delivery

Data must reach the **correct destination**.

2. Accuracy

Data must arrive **without errors**.

3. Timeliness

Data must arrive **within acceptable time**.

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6. Components of a Data Communication System

A communication system includes:

1. **Sender**
    - Device that sends data.
2. **Receiver**
    - Device that receives data.
3. **Message**
    - Information being transmitted.
4. **Medium**
    - Transmission path.
5. **Protocol**
    - Rules governing communication.

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7. Direction of Data Flow

Simplex

- One-direction communication only.

Example:

- TV broadcasting

Half-Duplex

- Both directions but **not simultaneously**.

Example:

- Walkie-talkies

Full-Duplex

- Both directions **simultaneously**.

Example:

- Telephone calls

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8. Access Networks

Access networks connect **end systems to the Internet**.

Types of Access Networks

1. Residential Access Networks

Used in homes.

Examples:

- Cable Internet
- DSL

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2. Institutional Access Networks

Used in organizations such as:

- Universities
- Companies

Technologies:

- Ethernet
- WiFi

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3. Mobile Access Networks

Used by mobile devices.

Examples:

- 4G
- 5G
- WiFi

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9. Cable Internet (HFC)

Hybrid Fiber Coaxial (HFC) network.

Characteristics

- Fiber connects ISP to neighborhoods
- Coaxial cable connects homes

Features

- Shared bandwidth
- Uses **Frequency Division Multiplexing (FDM)**

Typical speeds:

- Downstream: up to **1.2 Gbps**
- Upstream: **30–100 Mbps**

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10. DSL (Digital Subscriber Line)

Uses existing **telephone lines**.

Components

- DSL modem
- DSLAM (Digital Subscriber Line Access Multiplexer)

Characteristics

- Dedicated connection to ISP
- Data and voice transmitted at different frequencies.

Typical speeds:

- Downstream: **24–52 Mbps**
- Upstream: **3.5–16 Mbps**

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11. Home Networks

Typical home network components:

- Modem (DSL or Cable)
- Router
- Firewall
- WiFi access point
- Ethernet connections

Common speeds:

- Ethernet: **1 Gbps**
- WiFi: **54–450 Mbps**

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12. Physical Media

Physical media carry bits across the network.

Guided Media

Signals travel through physical cables.

Examples:

Twisted Pair

- Two copper wires twisted together
- Used in Ethernet

Speeds:

- Up to **10 Gbps**

Coaxial Cable

- Two concentric copper conductors
- Used in cable networks

Supports multiple channels.

Fiber Optic Cable

- Uses light signals
- Very high speed

Advantages:

- High bandwidth
- Low error rate
- Immune to electromagnetic interference

Speed:

- **10–100+ Gbps**

Unguided Media

Signals travel through air.

Examples:

- WiFi
- Cellular networks
- Bluetooth
- Satellite communication

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13. Packet Switching

Data is divided into **packets** before transmission.

Packets travel independently through the network.

Routers forward packets to their destination.

Store-and-Forward Transmission

Routers must receive the **entire packet** before forwarding it.

Transmission delay formula:

$$d_{trans} = \frac{L}{R}$$

Where:

- L = packet length (bits)
- R = transmission rate (bits/sec)

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14. Queueing

Packets may wait in a **queue** when the router is busy.

Occurs when:

Arrival rate > transmission rate.

Results:

- Delay

- Packet loss (if buffer full)

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15. Circuit Switching

Used in **traditional telephone networks**.

Characteristics:

- Dedicated communication path
- Reserved bandwidth
- No resource sharing

Examples:

- Telephone calls

Multiplexing Methods

Frequency Division Multiplexing (FDM)

- Different signals transmitted using different frequencies.

Time Division Multiplexing (TDM)

- Time divided into slots for each user.

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16. Packet vs Circuit Switching

Packet Switching

Advantages:

- Efficient resource sharing
- Suitable for bursty data

Disadvantages:

- Possible delay
- Possible packet loss

Circuit Switching

Advantages:

- Guaranteed bandwidth
- Predictable performance

Disadvantages:

- Inefficient resource use.

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17. Internet Structure

The Internet consists of multiple layers of ISPs.

Types of ISPs

1. **Access ISPs**
2. **Regional ISPs**
3. **Tier-1 ISPs**

Internet Exchange Points (IXP)

Facilities where ISPs connect to exchange traffic.

Content Provider Networks

Large companies operate their own networks.

Examples:

- Google
- Facebook
- Microsoft

Purpose:

- Deliver content faster to users.

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18. Network Performance

Important metrics:

- Delay
- Packet loss
- Throughput

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19. Types of Packet Delay

Total nodal delay:

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

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1. Processing Delay

Time to process packet header.

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2. Queueing Delay

Time waiting in router queue.

Depends on congestion.

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3. Transmission Delay

$$d_{trans} = \frac{L}{R}$$

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4. Propagation Delay

$$d_{prop} = \frac{d}{s}$$

Where:

- d = distance
- s = propagation speed

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20. Packet Loss

Occurs when router buffers become full.

Packet is dropped.

Recovery:

- Retransmission

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21. Throughput

Throughput = **rate at which data is successfully delivered**

Measured in:

- bits per second (bps)

Bottleneck Link

The **slowest link** determines the end-to-end throughput.

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22. Network Security

The Internet was not originally designed with strong security.

Common threats include:

Packet Sniffing

Attackers capture packets.

Tools:

- Wireshark

IP Spoofing

Attacker sends packets with a **fake source address**.

Denial of Service (DoS)

Attackers overwhelm a server with traffic.

Goal:

- Make services unavailable.

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23. Security Defenses

Authentication

Verify identity.

Confidentiality

Encryption protects data.

Integrity

Ensures data is not modified.

Access Control

Restricts network access.

Examples:

- Passwords
- VPN

Firewalls

Network security systems that filter traffic.

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24. Protocol Layering

To manage complexity, networks use **layers**.

Each layer performs specific functions.

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25. Internet Protocol Stack

The Internet uses a **5-layer architecture**.

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1. Application Layer

Supports applications.

Examples:

- HTTP
- DNS
- SMTP

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2. Transport Layer

Process-to-process communication.

Protocols:

- TCP
- UDP

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3. Network Layer

Routing packets.

Protocol:

- IP

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4. Link Layer

Communication between neighboring nodes.

Examples:

- Ethernet
- WiFi

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5. Physical Layer

Transmission of bits.

Examples:

- Electrical signals
- Optical signals

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26. Encapsulation

Each layer adds its own header.

Data units:

Application → **Message**
Transport → **Segment**
Network → **Datagram**
Link → **Frame**

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27. OSI Reference Model

7-layer model:

1. Application
2. Presentation
3. Session
4. Transport
5. Network
6. Data Link
7. Physical

Difference from Internet Model

Internet stack **does not include**:

- Presentation layer
- Session layer

Their functions are handled in the **application layer**.